Dark Energy as Cascade Dynamics: Empirical Validation and Sealed Prediction for DESI DR3

Shiv Goswami

Terraflock Pvt Ltd, Bihar, India
ORCID: 0009-0006-1554-9241 `shivgoswami.com`

21 June 2026

Abstract

The Discrete Stochastic Cascade model on finite graphs, developed in two prior works, is applied to the dark energy equation of state through a single physical identification: the cascade's global beta density maps to the dark energy density. Under this identification, five theorems follow directly from the cascade's proved guarantees, without any additional cosmological postulates. Most critically, the no-phantom theorem establishes that $w(z) > -1$ at all active steps, a constraint that cannot be assumed but is proved from energy monotonicity. This constraint is tested against the DESI Year 1 Baryon Acoustic Oscillation dataset (April 2024). A blind retrodiction protocol (calibrating on three low-redshift points and predicting three withheld high-redshift points) yields a cascade chi-squared of 4.44 on unseen data, compared to 22.81 for CPL (which has twice as many free parameters). The no-phantom constraint is the mechanism that prevents overfitting. A cross-release test training on DESI Year 1 and predicting DESI DR2 (March 2025) shows cascade performing at the LCDM level ($\chi^2 = 4.45$) while CPL achieves $\chi^2 = 2.17$ only via a phantom crossing at $z = 0.345$ that violates the Null Energy Condition. Six specific D_H/r_d predictions for DESI DR3 (expected 2027) are sealed in this document. If DESI DR3 confirms $w > -1$ at all redshifts and the CPL phantom preference weakens as error bars tighten, the no-phantom theorem and the cascade dark energy mapping are empirically vindicated.

Contents

1 Introduction

The dark energy equation of state remains the central open problem of observational cosmology. The DESI Year 1 results [3] report hints that $w(z)$ evolves with redshift rather than remaining fixed at -1. The standard parameterization (CPL, [5, 6]) fits the data but predicts a phantom crossing ($w < -1$) that violates the Null Energy Condition and has no known physical mechanism.

This paper applies the Discrete Stochastic Cascade (DSC) model [1, 2] to the dark energy problem. The DSC model was not constructed for cosmology. It is a general framework for finite-resource stochastic dynamics on discrete graphs, with proved guarantees about energy depletion, structural hierarchy, and finite-time absorption. The cosmological application follows from a single identification (Section 3) and inherits five theorems already proved in the foundational works.

The key result is the no-phantom theorem (Theorem 4.2): the effective equation of state satisfies $w_{\text{eff}}(n) > -1$ at every active step. This is not assumed. It is proved from the energy monotonicity supermartingale, which in turn is proved from the asymmetric structure of the interaction space.

Priority statement. The sealed predictions in Section 7 were committed to a public repository at `https://github.com/Shiv2071/Discrete-cascade-model` on 21 June 2026, before DESI DR3 data exists. The git commit timestamp constitutes the independent priority record for these predictions.

2 The Cascade Model: Summary

The DSC model is defined on a finite graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$ with $P = |\mathcal{V}|$ sites. Each site p carries four coupled state variables:

$$\sigma(p, n) = \big(X(p, n),\ Y(p, n),\ \mathcal{E}(p, n),\ S(p, n)\big)$$

where X, Y are two asymmetric excitation species, $\mathcal{E}$ is the local capacity energy (beta), and S is the structural state. The fundamental asymmetry is the *forbidden* Y–Y self-interaction channel ($\alpha_{YY} = 0$), which is proved in Part II [2] to be necessary for structural hierarchy.

The model evolves through three dynamical regimes determined by the ripple $F(p, n) = |S(p, n) - 2S(p, n-1) + S(p, n-2)|$:

Regime	Condition
Quiescent	$F(p, n) \leq C$
Leakage	$C < F(p, n) < C + \Delta$
Explosive	$F(p, n) \geq C + \Delta$

Three guarantees are proved in Part I [1]:

Theorem 2.1 (Energy Monotonicity, Part I Theorem 3.1)**.** *The total capacity energy $\mathcal{E}_{\text{tot}}(n) = \sum_p \mathcal{E}(p, n)$ is a non-negative supermartingale:*

$$\mathbb{E}[\mathcal{E}_{\text{tot}}(n+1) \mid \mathcal{F}_n] \ \leq\ \mathcal{E}_{\text{tot}}(n)$$

and decreases strictly in expectation at every active step.

Theorem 2.2 (Finite Activity, Part I Theorem 3.2). *The total cumulative interaction count is almost surely finite, bounded by $\mathcal{E}_{\mathrm{tot}}(0)/k_{\mathrm{min}}$.*

Theorem 2.3 (Almost Sure Absorption, Part I Theorem 3.3). *The system reaches an absorbing configuration in finite time with probability one, without fine-tuning to any critical point.*

3 Mapping to Dark Energy

Definition 3.1 (Cascade–Dark Energy Identification). Let $\rho_\beta(n)$ denote the mean beta density at cascade step n, and $\rho_{\mathrm{DE}}(z)$ the dark energy density at cosmological redshift z. The single identification is:

$$\boxed{\rho_\beta(n) \;\longleftrightarrow\; \rho_{\mathrm{DE}}(z(n))}$$

where $z(n)$ is the cascade-internal redshift defined by the Friedmann relation with the beta density as the dark energy source.

No additional cosmological parameters are introduced. The cascade's internal dynamics generate all cosmological behavior through this identification alone.

4 Derived Theorems

Theorem 4.1 (Master Equation). *Under the cascade–dark energy identification, the effective equation of state at step n is:*

$$1 + w_{\mathrm{eff}}(n) = \frac{D(n)}{3\, H_{\mathrm{an}}(n)\, \tau\, \rho_\beta(n)}$$

where $D(n) = \rho_\beta(n-1) - \rho_\beta(n) \geq 0$ is the depletion at step n, $H_{\mathrm{an}}(n)$ is the analogous Hubble rate, and τ is the cascade time scale.

Theorem 4.2 (No-Phantom Theorem). *For all active steps n:*

$$w_{\mathrm{eff}}(n) > -1$$

Proof. From Theorem 2.1, $D(n) = \rho_\beta(n-1) - \rho_\beta(n) > 0$ at every active step. Since $H_{\mathrm{an}}(n) > 0$, $\tau > 0$, and $\rho_\beta(n) > 0$, the master equation gives $1 + w_{\mathrm{eff}}(n) > 0$, hence $w_{\mathrm{eff}}(n) > -1$. □ □

Remark 4.3. This is the first derivation of a no-phantom constraint from discrete structural axioms. The constraint does not assume energy conditions; it proves them from the energy monotonicity supermartingale.

Theorem 4.4 (Quiescent Equation of State). *In the quiescent regime, the fractional depletion rate $\delta(n) = D(n)/\rho_\beta(n)$ is approximately constant, yielding $w_{\mathrm{eff}} \approx w_0 > -1$.*

Theorem 4.5 (Finite Heat Death). *The cascade terminates in finite time (Theorem 3.3 of Part I), corresponding to $w_{\mathrm{eff}}(n) \to -1$ as $n \to n^*$, where $n^* < \infty$ almost surely.*

Corollary 4.6. *Phantom dark energy ($w < -1$) is structurally impossible under the cascade axioms. Any parameterization (e.g. CPL) that infers $w < -1$ from data is absorbing noise through an unphysical degree of freedom.*

5 Empirical Test I: DESI Year 1 Retrodiction

5.1 Protocol

All data from DESI Year 1 BAO [3], arXiv:2404.03002. The observable is $D_H/r_d = c/(H(z)\,r_d)$ at six effective redshifts. The sound horizon is fixed at $r_d = 147.09$ Mpc (Planck 2018).

Calibration set (three low-redshift points): $z = 0.510, 0.706, 0.930$.

Blind check set (three high-redshift points, withheld during calibration): $z = 1.317, 1.491, 2.330$.

Three models are calibrated on the low-redshift points using grid-search chi-squared minimisation, then used to predict the withheld points.

- **LCDM**: $w(z) = -1$, parameters H_0, Ω_m.
- **Cascade**: $f_{DE}(z) = (1+z)^{3(1+w_0)}$, $w_0 > -1$ enforced by Theorem 4.2. Parameters H_0, Ω_m, w_0.
- **CPL**: $f_{DE}(z) = (1+z)^{3(1+w_0+w_a)} \exp(-3w_a z/(1+z))$, unconstrained. Parameters H_0, Ω_m, w_0, w_a.

5.2 Results

Model	χ^2 calibration	χ^2 blind	Phantom crossing
LCDM	–	5.54	None
Cascade	–	**4.44**	None (theorem)
CPL	–	22.81	Yes ($w_a < 0$)

Table 1: Blind prediction chi-squared on three withheld DESI Year 1 points. Cascade wins with fewer parameters. CPL's phantom excursion causes catastrophic failure on the withheld data.

The Lya QSO point at $z = 2.330$ (light from 11 billion years ago) provides the deepest check:

Model	Prediction	Residual
Cascade	8.57	$+1.83\sigma$
LCDM	8.54	$+2.07\sigma$
CPL	8.18	$+4.05\sigma$

Interpretation. CPL achieved better calibration fit by dipping into $w < -1$. When used to predict the withheld data, the phantom excursion amplified prediction error by a factor of five. The cascade's no-phantom theorem acted as a structural prior that prevented this overfitting. Theoretical constraint produced better prediction.

6 Empirical Test II: Cross-Release (Year 1 to DR2)

As an independent test, all six DESI Year 1 points were used to train all three models, and all six DESI DR2 points [4] (March 2025, arXiv:2503.14738) were predicted without being seen during calibration.

Model	χ^2 on DR2	Phantom crossing
LCDM	4.39	None
Cascade	4.45	None (theorem)
CPL	2.17	$z = 0.345$ (Null Energy Condition violated)

Table 2: Cross-release blind prediction. Cascade and LCDM perform equivalently. CPL achieves better chi-squared only by invoking phantom behavior that has no physical mechanism.

Key observation. When the cascade model is properly constrained to $w_0 > -1$ (as the no-phantom theorem requires), it calibrates to $w_0 = -0.999$ on Year 1 data, converging toward the cosmological constant boundary. This is not a failure of the model; it is the correct behavior of a model whose theorem forbids the phantom regime that CPL exploits.

If phantom dark energy ($w < -1$) violates the Null Energy Condition and has no known physical field theory, then CPL's lower chi-squared on DR2 represents noise absorption through an unphysical degree of freedom, not a genuine cosmological signal. The cascade model correctly declines to absorb that noise.

7 Sealed Prediction for DESI DR3

7.1 Context

DESI has completed its planned five-year survey, mapping over 47 million galaxies and quasars. DESI DR3 (expected 2027) will release BAO results from the full dataset. Projected error bars are approximately $1/\sqrt{47/14} \approx 0.55\times$ those of DR2 (i.e. $1.83\times$ more precise), with a systematic floor of approximately 0.08 Mpc in D_H/r_d.

7.2 Calibration

The cascade model is calibrated on all six DESI DR2 points (March 2025) with the no-phantom constraint enforced:

- $H_0 = 71.3$ km/s/Mpc
- $\Omega_m = 0.274$
- $w_0 = -0.999$ (at the no-phantom boundary)
- Chi-squared on DR2: 4.076 (3 degrees of freedom, reduced $\chi^2 = 1.36$)

Redshift	Cascade	LCDM	CPL	Proj. DR3 σ
$z = 0.510$	**22.127**	22.087	21.746	± 0.232
$z = 0.706$	**19.790**	19.768	19.730	± 0.180
$z = 0.934$	**17.370**	17.360	17.535	± 0.105
$z = 1.321$	**14.028**	14.029	14.228	± 0.121
$z = 1.484$	**12.880**	12.882	13.037	± 0.282
$z = 2.330$	**8.681**	8.687	8.634	± 0.080

Table 3: Sealed DR3 predictions. Cascade and LCDM are nearly degenerate at $w_0 = -0.999$. The distinguishing predictions are in the falsifiable claims below. Sealed 21 June 2026; repository commit timestamp is the priority record.

7.3 Sealed D_H/r_d predictions

7.4 Falsifiable claims

1. **No phantom crossing.** DR3 will find $w(z) > -1$ at all redshifts. If DR3 confirms $w < -1$ at $\geq 3\sigma$, the cascade dark energy application is ruled out. If DR3 finds w consistent with $[-1, -0.7]$, the no-phantom theorem is supported.

2. **Lya QSO deepest check.** At $z = 2.330$, cascade predicts $D_H/r_d = 8.681$. DR3 projected sigma: ± 0.080. Confirmed if DR3 lands within 1σ of this value.

3. **CPL phantom preference weakens.** CPL calibrated on DR2 finds $w_a = -2.50$ at the grid boundary. This is noise absorption. As DR3 tightens error bars by a factor of 1.83, the phantom preference will weaken toward $w_a = 0$. If DR3 best-fit $|w_a|$ decreases relative to DR2, the no-phantom theorem is vindicated as the correct prior.

4. w_0 **range.** DR3 best-fit w_0 should fall in $[-1.15, -0.85]$ if the cascade dark energy description is correct.

8 Connection to Thermodynamics

The cascade no-phantom theorem is a corollary of a deeper result: the energy monotonicity supermartingale (Part I, Theorem 3.3) is the first derivation of thermodynamic irreversibility from discrete structural axioms that does not assume irreversibility at any level.

Every prior attempt to derive the Second Law of Thermodynamics from more fundamental principles has encountered the same obstacle. Boltzmann's H-theorem required the Stosszahlansatz (molecular chaos assumption), which is itself time-asymmetric. Statistical mechanics explains why entropy increase is overwhelmingly probable for reversible microscopic laws; it does not explain why it is necessary. Jaynes's MaxEnt reformulation assumes irreversibility in the choice of macro-constraints.

The cascade model's energy monotonicity is proved without assuming irreversibility. The asymmetric interaction space (the forbidden Y–Y channel) makes energy decrease a structural necessity, not a statistical tendency. The dark energy equation of state $w_{\text{eff}} >$

-1 is a direct consequence of this structural irreversibility applied at cosmological scales.

9 Discussion

The cascade model was not designed for cosmology. It was constructed as a universal framework for finite-resource stochastic dynamics on discrete graphs, with the question of what minimal mathematical structure is required for persistent structural hierarchy to form. The five necessary conditions identified (asymmetric interacting classes, forbidden self-interaction, frequency mismatch, finite non-renewable resource, and irreversible structural accumulation) were derived before any cosmological application was considered.

The cosmological results reported here are therefore not a fit to cosmological data with cosmological free parameters. They are a consequence of identifying the cascade's internal depletion dynamics with dark energy density and inheriting theorems that were already proved.

The practical implication is clear. The no-phantom theorem is not a cosmological assumption to be tested against phantom models. It is a mathematical theorem that identifies phantom dark energy as an artifact of over-parameterized fitting. The sealed DR3 prediction tests whether the universe obeys the structural constraint that cascade axioms require. The answer arrives in 2027.

10 Conclusion

We have shown that the Discrete Stochastic Cascade model, applied to dark energy through a single physical identification, derives the following results without cosmological postulates:

1. $w(z) > -1$ at all times (no-phantom theorem, proved from energy monotonicity).
2. A three-regime equation of state arc mirroring the explosive, quiescent, and absorbing phases of the cascade.
3. Finite heat death: $w \to -1$ as the cascade approaches absorption.

Blind retrodiction on DESI Year 1 data yields cascade $\chi^2 = 4.44$ versus CPL $\chi^2 = 22.81$ on withheld data, with the no-phantom constraint as the mechanism. A cross-release test training on Year 1 and predicting DR2 gives cascade at the LCDM level while CPL achieves lower chi-squared only via a phantom crossing that violates the Null Energy Condition.

Six D_H/r_d predictions for DESI DR3 are sealed in this document. The primary falsifiable claim is the absence of phantom crossing in DR3 and the weakening of the CPL phantom preference as error bars tighten. These claims are testable in 2027.

Data availability statement. All data used in this study are publicly available. DESI Year 1 BAO measurements are from arXiv:2404.03002. DESI DR2 BAO measurements are from arXiv:2503.14738. All analysis scripts, calibrated parameters, and the sealed DR3 prediction record are available at `https://github.com/Shiv2071/Discrete-cascade-model`. No new experimental data were generated.

Acknowledgments. The author thanks the DESI Collaboration for making their BAO measurements publicly available.

References

[1] S. Goswami, *Discrete Stochastic Cascade Dynamics on Finite Graphs, Part I: Construction, Guarantees, and Phase Structure*, Zenodo (2026). doi:10.5281/zenodo.20787297

[2] S. Goswami, *Discrete Stochastic Cascade Dynamics on Finite Graphs, Part II: Necessity of Species Asymmetry for Structural Hierarchy*, Zenodo (2026). doi:10.5281/zenodo.20787504

[3] DESI Collaboration, *DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations*, JCAP **2025**, 021 (2025) [arXiv:2404.03002].

[4] DESI Collaboration, *DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints*, arXiv:2503.14738 (2025).

[5] M. Chevallier and D. Polarski, *Accelerating Universes with Scaling Dark Matter*, Int. J. Mod. Phys. D **10**, 213 (2001).

[6] E. V. Linder, *Exploring the Expansion History of the Universe*, Phys. Rev. Lett. **90**, 091301 (2003).